

ICD-11 PTSD, Complex PTSD, and Substance Use Disorder: Associations and Clinical Correlates

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Objective: Trauma- and substance-related disorders are highly comorbid. The co-occurrence of these conditions is associated with a more severe clinical profile in patients, and this remains a significant challenge for treatment providers. Previous research examining this comorbidity has focused on *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition, posttraumatic stress disorder (PTSD), resulting in a significant lack of research examining other formulations of trauma-related disorders, in particular complex posttraumatic stress disorder (CPTSD). It remains unclear whether CPTSD requires additional treatment approaches in substance use treatment settings. The present study sought to expand previous research by examining the *International Classification of Diseases 11th revision* PTSD and Complex PTSD (CPTSD) diagnoses in a substance use treatment setting, and to explore the associations between symptoms of PTSD/CPTSD and characteristics of substance use disorder. **Method:** The study utilized data collected from individuals commencing substance use treatment ($n = 72$). Assessments included trauma exposure, trauma-related symptoms, primary substance of concern, psychological dependence, and recent substance use. **Results:** Forty percent of the sample screened positive for a trauma-related disorder, with CPTSD more prevalent (34.8%) than PTSD (5.8%). There was limited evidence of associations between trauma-related symptoms and substance use characteristics. Severity of each CPTSD symptom cluster (affect dysregulation, negative self-concept, disturbances in relationships) was significantly, positively associated with psychological dependence, a pattern not found for PTSD symptoms. **Conclusions:** Given that CPTSD is more prevalent and demonstrates differential relationships to substance use characteristics, future research should examine whether individuals with comorbid CPTSD would benefit from adjusted treatment approaches.

Clinical Impact Statement

Individuals presenting for substance use treatment demonstrate high levels of exposure to traumatic events, and an increased prevalence of trauma-related disorders. Complex posttraumatic stress disorder (CPTSD) is more prevalent than PTSD in this population. There appear to be differential associations between symptoms of PTSD versus CPTSD and psychological dependence to a substance. The additional symptoms associated with CPTSD should be considered in treatment—including via direct screening at assessment. Existing, evidence-based treatments should be adapted to address the additional CPTSD-related symptoms within substance use treatment.

Keywords: posttraumatic stress disorder, complex posttraumatic stress disorder, trauma, substance-related disorders, addiction

This article was published Online First June 12, 2025.

Tyson D. Bailey served as action editor.

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The authors report that there are no conflicts of interest to declare. Logan R. Harvey is supported by a Postgraduate Research Scholarship in Substance Use Disorders from the Matilda Centre, University of Sydney. Katherine L. Mills is supported by an Australian Government National Health and Medical Research Council Investigator Grant. Christina Marel is supported by Australian Government National Health and Medical Research Council Translating Research into Practice Fellowship. The authors acknowledge the patients and staff of the Drug Health Service, Western Sydney Local Health District for their important contributions to this study.

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Logan R. Harvey played a lead role in conceptualization, data curation, formal analysis, investigation, methodology, project administration, writing—original draft, and writing—review and editing. Tim Slade played a supporting role in conceptualization, formal analysis, methodology, and writing—review and editing and an equal role in supervision. Christina Marel played a supporting role in conceptualization, formal analysis, methodology, and writing—review and editing and an equal role in supervision. Katherine L. Mills played a supporting role in conceptualization, formal analysis, methodology, and writing—review and editing and an equal role in supervision.

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For individuals presenting for treatment of substance use disorder (SUD), trauma-related disorders are a prevalent comorbidity. In Australia, it is estimated that between 5 and 66% of those presenting for substance use treatment will meet criteria for current posttraumatic stress disorder (Kingston et al., 2017) with the predominance of studies deriving estimates that are considerably higher than those observed in the general population (4.4%; McEvoy et al., 2011). Individuals presenting with this comorbidity are more likely to experience additional comorbid mental and physical health concerns (Bowe & Rosenheck, 2015; Prior et al., 2017), greater severity of SUD (Patel et al., 2021), and more significant functional and psychosocial impairment (L. R. Cohen & Hien, 2006; Drapkin et al., 2011). Consistent with this, PTSD has been identified as a key treatment challenge for substance use treatment providers (Gielen et al., 2014; Killeen et al., 2015; Mefodeva et al., 2023; Roberts et al., 2023).

The co-occurrence of SUD and PTSD has most commonly been understood in terms of the self-medication hypothesis—the theory that substance use functions to relieve symptoms of PTSD (Khantzian, 1997). In line with this, symptoms of PTSD and SUD are mutually reinforcing (Tripp et al., 2020), and individuals with this comorbidity experience cravings in response to both trauma- and substance-related cues (Coffey et al., 2002). Individuals with co-occurring PTSD and SUD experience worse clinical outcomes compared with those with single diagnoses, particularly in relation to functional and psychosocial impairments (L. R. Cohen & Hien, 2006; Najt et al., 2011). Effective treatments do exist; however, these demonstrate high levels of attrition (Hien et al., 2023; Roberts et al., 2022). These treatments typically involve the parallel or integrated targeting of both PTSD- and SUD-related symptoms. In general, trauma-focused interventions are effective in the presence of co-occurring substance use and there is evidence that reductions in trauma-related symptoms correspond to reductions in SUD-related symptoms—highlighting the significance of targeting this comorbidity within SUD treatment (Back et al., 2014; Hien et al., 2010, 2018; Kaczurkin et al., 2016; Roberts et al., 2023). The available literature indicates that pathways to recovery from substance use are often nonlinear, and there have been limited findings to date which help identify individuals who might be treatment responders versus those who do not (López-Castro et al., 2015). Given the increased symptom severity and high rates of treatment attrition in this presentation, it remains important to further understand the relationship between PTSD and SUD to inform refinements to treatment options.

The majority of studies that have examined this relationship have focused on hyperarousal symptoms. These studies have found positive associations between arousal/hyperarousal symptoms and measures of substance use including age of onset, duration of SUD, number of substances used, patterns of recent substance use, and pharmacotherapy dosage (Dell’Osso et al., 2014; Love & Torgerson, 2019; Shipherd et al., 2005; Stewart et al., 1999; Tull et al., 2010). In a systematic review of the comorbidity between PTSD and alcohol misuse, Debell et al. (2014) reported a consistent positive association between arousal/hyperarousal and alcohol misuse, and other studies have suggested a similar relationship for anxiolytics, analgesics (Stewart et al., 1999), cocaine, and cannabis (Khouri et al., 2010), but not for crack/cocaine (Tull et al., 2010) or heroin (Khouri et al., 2010; Tull et al., 2010). Symptoms of avoidance and numbing have been associated with broad categories of alcohol and other drug use, albeit with even less consistency in results (Boland et al., 2018; Love &

Torgerson, 2019; McFall et al., 1992). Regarding specific substances, symptoms of avoidance/numbing were identified in the Debell et al.’s (2014) review as consistently associated with alcohol misuse, while positive associations have also been identified with anxiolytic use (Stewart et al., 1999), cocaine use, and cannabis use (Khouri et al., 2010). In contrast, Tull et al. (2010) did not identify a relationship between crack/cocaine use and avoidance/numbing symptoms but did identify a negative association between avoidance/numbing and heroin use. Negative alterations in cognitions and mood were found to be associated with alcohol use in one study of U.S. veterans (Boland et al., 2018), but this result was not replicated in a nonclinical sample which similarly did not identify a relationship between a broad “other substances” (i.e., nonalcohol) category and negative alterations in mood and thinking (Love & Torgerson, 2019). Finally, reexperiencing symptoms have been found to be associated with alcohol (Debell et al., 2014; Love & Torgerson, 2019) and other drug use (McFall et al., 1992) including cocaine and cannabis (Khouri et al., 2010), and analgesics (Stewart et al., 1999).

Importantly, the relationships identified in these studies have not been consistently replicated across studies. This may be due in part to differential endorsement of individual symptoms across different samples but also to the inclusion of different measures of different iterations of the *Diagnostic and Statistical Manual of Mental Disorders (DSM)* PTSD symptom formulations across time (e.g., *Diagnostic and Statistical Manual of Mental Disorders*, Third Edition, to *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition, Text Revision). As diagnostic systems have developed, further research is required to expand current knowledge. In addition to PTSD, the *International Classification of Diseases 11th revision (ICD-11; World Health Organization, 2019)* has included a new diagnosis of Complex PTSD (CPTSD), which provides an opportunity to reexamine the above relationships but also expand these to the new CPTSD diagnosis. This is important also as there is a significant lack of research examining the relationship between ICD trauma-related disorders and SUDs in general. Complex PTSD in this formulation requires an individual to first meet the criteria for PTSD (reexperiencing symptoms, avoidance symptoms, and persistent sense of threat) but also to demonstrate the experience of three other symptom clusters reflecting *disturbances in self-organization* (affect dysregulation, negative self-concept, and difficulties in relationships). Given this, it is possible that the additional symptom clusters may help further explain the relationship between trauma-related pathology and SUD, including in treatment. For example, difficulties managing and regulating emotions (as per the CPTSD symptom cluster of emotion regulation difficulties) have been found to be associated with less substance use treatment completed and to also be a maintaining factor in substance use disorder (Cheetham et al., 2010; Tull et al., 2013).

The ICD-11 CPTSD diagnosis has been found to most commonly proceed the experience of cumulative, childhood, or interpersonal trauma (Frost et al., 2019; Maercker et al., 2022; Palic et al., 2016; Powers et al., 2017) and is associated with higher levels of symptom severity, functional impairment, lower levels of psychological well-being, and greater levels of co-occurring mental health problems than the PTSD diagnosis (Bressington et al., 2024; Cloitre et al., 2019; Howard et al., 2021; Murphy et al., 2021). This includes higher levels of loneliness, sleep problems, use of psychotropic medications, and suicidality (Folke et al., 2023; McGinty et al., 2024). Importantly, these patterns of trauma exposure and comorbidity are commonly

seen among people with SUD (Diehl et al., 2019; Ruglass et al., 2017; Santo et al., 2021). Within psychiatric clinical populations (compared with community samples), there is a higher prevalence of CPTSD than PTSD (Maercker et al., 2022). For example, in an Australian sample of treatment seeking veterans, Bressington et al. (2024) identified significantly higher rates of CPTSD versus PTSD (51.4% vs. 9.1%). However, the prevalence and clinical correlates of CPTSD among people in substance use treatment services have yet to be examined. Early indicators suggest that there is a strong comorbidity between CPTSD and SUD. Powers et al. (2017) found that *Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition, Text Revision, alcohol abuse and dependence were more prevalent in an *ICD-11* CPTSD group than a *DSM-5* PTSD group, although Karatzias, Hyland, et al. (2019) found similarly high rates of AUD in individuals with *ICD-11* PTSD and CPTSD (60.0% vs. 58.8%). Finally, Facer-Irwin et al. (2022) found that those with *ICD-11* PTSD were more likely to screen positive for alcohol dependence ($OR = 3.2$), while those with *ICD-11* CPTSD were more likely to screen positive for other substance use ($OR = 3.4$). It is difficult to draw conclusions from these results, however, as only a small number of studies exist and they compare different diagnostic systems. There is emerging evidence that existing treatments for PTSD may have an effect on symptoms of CPTSD (Karatzias, Murphy, et al., 2019), however, no studies to date have evaluated the treatment of comorbid CPTSD–SUD, and it is unclear whether the existing literature related to PTSD can be generalized to this group. Importantly also, for existing PTSD–SUD treatments, clinically meaningful improvements are only demonstrated for approximately half of those who undertake them (Mills et al., 2016). Given this and the fact that the majority of research to date has focused on the core symptoms of PTSD, it remains unclear whether the additional symptom clusters associated with CPTSD will further understandings of treatment options. Based on existing research in other clinical populations, it is likely that CPTSD will be prevalent in this population, and it is therefore crucial to evaluate the relationship between CPTSD and SUD.

The Present Study

The aim of the present study then, was to expand previous research by first examining the prevalence of *ICD-11* PTSD and CPTSD in a substance use treatment setting, and second to explore the associations between *ICD-11* trauma-related symptom clusters and characteristics of SUD (primary substance of concern, recent substance use, and psychological dependence). As most previous research has focused on substance use behavior or on specific substances, the present study will additionally examine psychological dependence to a substance. Psychological dependence is an important characteristic of substance use disorders and reflects the nonphysiological components of dependence to a psychoactive substance (Raistrick et al., 1994). This component of dependence is reflected in diagnostic formulations for substance-related conditions across both *DSM* and *ICD* systems.

Materials and Method

Design

The present study analyzed data from the *Trauma and Substance Use in Treatment* study, a prospective cohort study of 72 people

recruited from a large, public metropolitan substance use treatment service based in greater Sydney, Australia. Ethics approval was obtained from the Western Sydney Local Health District Human Research Ethics Committee (HREC/18/WMEAD/441).

Participants were recruited as they entered substance use treatment. All eligible patients entering treatment with a counselor or psychologist for substance use issues were invited to participate.

Eligibility criteria were as follows:

(a) aged 18 years or over; (b) Australian citizen or permanent resident; (c) fluent in English; and (d) agreed to provide contact details for follow-up.

Measures

Participants were administered a series of measures by their treating clinician in their initial clinical assessment sessions. These measures assessed demographics, substance use (past month), trauma exposure (lifetime), and trauma symptoms (past month).

Demographics

Demographics included participant age (years) and gender (“male,” “female,” “transgender,” “do not identify as female, male or transgender,” or “prefer not to say”).

Substance Use

Substance use characteristics were assessed as part of routine data collection upon treatment entry using the NSW Health Drug and Alcohol Clinical Assessment and the Australian Treatment Outcomes Profile (Ryan et al., 2014). This included self-identified primary substance of concern and secondary substance of concern (categorized as “alcohol,” “cannabis,” or “other illicit substance,” the latter group consisting of cocaine, heroin, and methamphetamine); and the number of days in the past month they had used their primary substance (maximum of 28 days). The Leeds Dependence Questionnaire (LDQ; Raistrick et al., 1994) was used to assess psychological dependence in relation to primary substance of concern. The LDQ is a 10-item self-report measure of psychological dependence to psychoactive substances. Scores range from 0 to 30, with higher scores indicating a greater level of psychological dependence. The LDQ has been found to have sound psychometric properties, including reliability, validity, and sensitivity to change (Heather et al., 2001; Raistrick et al., 2014). In the present study, the LDQ demonstrated sound reliability ($\alpha = .93$).

Trauma, PTSD, and CPTSD

The Childhood Trauma Questionnaire (CTQ) was used to assess trauma exposure specifically in childhood (Bernstein & Fink, 1998). The CTQ is a 28-item measure that assesses exposure to five domains of childhood maltreatment: physical abuse, sexual abuse, emotional abuse, physical neglect, and emotional neglect. Scores are summed for each domain, with higher scores reflecting more severe levels of maltreatment. Endorsement of each domain of childhood maltreatment was calculated based on the suggested cut scores for the measure (Bernstein & Fink, 1998). The Life Events Checklist for *DSM-5* (LEC5; Weathers et al., 2013) was administered to assess trauma exposure more broadly. In considering

exposure to LEC-5—only those *directly experienced* were considered for this particular study. The International Trauma Questionnaire (ITQ; Cloitre et al., 2018) was used to assess symptoms of ICD-11 PTSD and CPTSD. Participant responses to the ITQ were scored as per the recommended diagnostic algorithm (Cloitre et al., 2018). Specifically, participants who endorsed an individual symptom at “moderately” or higher on at least one item assessing the symptom were considered to endorse the symptom. For participants to screen positive for ICD-11 PTSD, they were required to endorse one or more symptoms from each of the three PTSD core symptom clusters (re-experiencing, avoidance, and persistent sense of threat) and functional impairment related to those symptoms. For ICD-11 CPTSD, participants were required to meet criteria for the PTSD diagnosis and additionally endorse one or more symptoms from each of the three CPTSD symptom clusters (affect regulation issues, negative self-concept, and interpersonal issues) and functional impairment related to those symptoms. A summed score for each symptom cluster was calculated to act as a measure of symptom severity. The ITQ has been validated in a range of populations and settings and demonstrates sound psychometric properties (Redican et al., 2021). In the present study, the ITQ demonstrated sound reliability for the full scale ($\alpha = .94$) PTSD scale ($\alpha = .82$) and CPTSD subscale ($\alpha = .88$).

Data Analysis

There was a small amount (<1%) of missing data across the survey questions. Numbers are reported alongside proportions for categorical variables broken down by group for demographic, substance use, and trauma variables. The distributions of all continuous variables to be examined inferentially were found to deviate from normality, with significant issues with skewness evident as assessed with the Shapiro-Wilk and Kolmogorov-Smirnov tests and by visual inspection of Q-Q plots. Medians are reported with interquartile ranges (IQR) for continuous variables. Inferential analyses were carried out using median scores primarily, however, means and standard deviations are additionally reported for key study variables to allow comparison to the broader literature.

To evaluate the associations between primary substance groups and the trauma-related symptom clusters, Kruskal-Wallis and Mann-Whitney tests were utilized. For the Kruskal-Wallis test, effect sizes were calculated using η -squared calculated for the H statistic (η^2 ; B. H. Cohen, 2008; Tomczak & Tomczak, 2014). This effect size reflects the variance in mean rank accounted for by the group variable, analogous to the eta-squared used within a simple analysis of variance (B. H. Cohen, 2008). These effect sizes were interpreted with the available convention for the standard η^2 based on the f statistics, where small = .01, medium = .06, and large = .14 (J. Cohen, 1988). Pairwise comparisons were then conducted with Mann-Whitney U tests utilizing a Bonferroni correction for multiple comparisons, with Glass's rank-biserial correlation (r_G) used to estimate effect sizes (Glass, 1966; Tomczak & Tomczak, 2014). This effect size provides a measure of the difference in average rank between two groups and is similar to the Pearson r statistic (B. H. Cohen, 2008). These effect sizes were interpreted based on the available conventions, with small = .1, medium = .3, and large = .5 (J. Cohen, 1988). To assess associations between symptom clusters and the other substance use characteristics (recent substance use, psychological dependence), correlation coefficients were calculated using Spearman's rank-order correlation (r_s).

Results

Sample Characteristics

Sixty-one percent of the sample identified as male ($n = 44$, 61.1%) with the remainder identifying as female ($n = 28$, 38.9%). The mean age was 35.78 ($SD = 10.09$). Half the sample had received prior substance use treatment ($n = 40$, 55.6%).

Alcohol was the most common primary drug of concern ($n = 29$, 40.3%); cannabis was endorsed as the primary drug of concern for 20.8% ($n = 15$). Thirty-nine percent ($n = 28$) endorsed other illicit drugs as their primary drug of concern, predominantly methamphetamine ($n = 26$, 36.1%). Other drugs included in this category were cocaine ($n = 1$; 1.4%) and heroin ($n = 1$, 1.4%). Just over one third of the sample also identified a secondary substance of concern ($n = 25$, 34.7%). Most participants reported using their primary substance in the 28 days preceding the assessment (70.3%), with the median number of days of recent substance use being 7.5 (IQR = 0.00, 22.50).

Trauma Exposure

One hundred percent of the sample had experienced at least one domain of childhood maltreatment (median of 3.5 different types IQR = 2.00, 4.00) and 94.6% had been exposed to a traumatic event as measured by the LEC5. The most prevalent domain of childhood maltreatment was physical neglect ($n = 70$, 98.6%), followed by emotional abuse ($n = 50$, 70.4%), emotional neglect ($n = 47$, 67.1%), physical abuse ($n = 41$, 57.7%), and sexual abuse ($n = 24$, 33.8%). The median number of LEC5 events experienced was 4.00 (IQR = 2.00, 5.75). Rates of exposure to events on the LEC5 are presented in Table 1.

Prevalence of PTSD and CPTSD Symptoms in Individuals Presenting for Substance Use Treatment

Prevalence of endorsed ICD-11 PTSD and CPTSD symptom clusters is presented in Table 2. Each symptom cluster of PTSD and

Table 1

Trauma Exposure—Life Events Checklist for Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition

Event	<i>n</i>	%
Physical assault	48	66.7
Transportation accident	39	54.2
Other stressful event	39	54.2
Assault with a weapon	24	33.3
Sexual assault	23	32.4
Unwanted/uncomfortable sexual experience	22	30.6
Serious accident at work/home/recreational activity	18	25.0
Fire or explosion	16	22.2
Life-threatening illness/injury	16	22.2
Injury/harm to other	16	22.2
Natural disaster	11	15.3
Exposure to toxic substance	8	11.1
Captivity	7	9.7
Sudden violent death	7	9.7
Sudden accidental death	7	9.7
Severe human suffering	6	8.3
Combat/war zone exposure	4	5.6

Table 2
ICD-11 Trauma Symptoms—International Trauma Questionnaire

Diagnosis and symptom cluster	<i>n</i>	%
PTSD	4	5.8
Reexperiencing	38	52.8
Avoidance	53	74.6
Persistent sense of threat	49	70.0
PTSD-related functional impairment	48	66.7
CPTSD	24	34.8
Difficulties in relationships	58	80.6
Affect dysregulation	61	84.7
Negative self-concept	50	69.4
CPTSD-related functional impairment	56	77.8

Note. ICD-11 = International Classification of Diseases 11th revision; PTSD = posttraumatic stress disorder; CPTSD = complex posttraumatic stress disorder.

CPTSD was individually endorsed by at least 50% of the sample. Overall, 40.6% of the sample screened positive for either PTSD or CPTSD (5.8% and 34.8%, respectively).

Associations Between Trauma-Related Symptoms and Substance Use Characteristics

Tables 3 and 4 presents the median ITQ symptom cluster scores, across primary substance groups. The results of group comparisons for primary substance are presented from the Kruskal–Wallis test, along with pairwise comparisons utilizing the Mann–Whitney *U* test.

No statistically significant group differences were identified regarding the ITQ symptom clusters, and all effect sizes were small. Exploratory pairwise comparisons were similarly nonsignificant, although medium sized effects were found when comparing the alcohol and cannabis groups in regard to each functional impairment symptom cluster and the disturbance in relationships symptom cluster.

Table 5 presents the correlation matrix for the remaining substance use variables (psychological dependence and recent substance use) and ITQ symptom cluster scores. There was no evidence of association between any of the ITQ symptom cluster scores and recent substance use. There were relationships identified between ITQ CPTSD symptom clusters and psychological dependence, with

significant, medium sized associations identified between psychological dependence and affect dysregulation, $r_s(68) = .35, p = .003$, negative self-concept, $r_s(68) = .38, p = .002$, disturbances in relationships, $r_s(66) = .43, p < .001$, and CPTSD-related functional impairment, $r_s(68) = .43, p \leq .001$.

Discussion

This study was one of the first to examine the prevalence and clinical correlates of ICD-11 PTSD and CPTSD in a population seeking substance use treatment. As expected, the sample demonstrated high rates of exposure to traumatic experiences across the lifespan. Unsurprisingly, there were correspondingly high rates of trauma-related symptoms identified. Rates of endorsement for each of the ICD-11 PTSD and CPTSD symptoms were above 50%, and over 40% of the sample screened positive for either trauma-related disorder. Of specific importance, rates of CPTSD were found to be considerably higher than PTSD. There are no previously published rates of ICD-11 PTSD or CPTSD in individuals presenting for substance use treatment by which the current results could be better understood. Broadly, the overall prevalence of 40.6% of the sample screening positive for either disorder is in line with earlier prevalence rates of PTSD in comparable samples (Kingston et al., 2017). The finding that the majority of these individuals (85.7%) reflected a CPTSD rather than a PTSD profile is significant and suggests that CPTSD is a particularly salient clinical issue in substance use treatment. This finding is consistent with other studies conducted among clinical samples. For example, previous research in individuals seeking trauma-related treatment at a public psychiatric service in Scotland found that 75.6% of individuals reflected a CPTSD profile, and only 24.4% endorsed a PTSD profile (Karatzias et al., 2017). The rates of CPTSD symptoms present in the current sample provide an important indicator for clinicians that assessment of posttraumatic psychopathology needs to be expanded beyond the well-established traditional symptoms of standard PTSD.

Although confirming the expectedly high rates of CPTSD in this sample, this study found limited evidence of differences between users of particular substances in regard to symptoms of PTSD and CPTSD. There was indication of a trend toward higher levels of PTSD and CPTSD-related functional impairment and problems in relationships in those who nominated alcohol rather than cannabis as their main drug of concern. This could suggest that there may be

Table 3
ITQ Symptom Cluster Score Descriptives Across Primary Drug of Concern Group

ITQ symptom cluster	Alcohol (1)		Cannabis (2)		Other (3)	
	<i>Mdn</i> (IQR)	<i>M</i> (<i>SD</i>)	<i>Mdn</i> (IQR)	<i>M</i> (<i>SD</i>)	<i>Mdn</i> (IQR)	<i>M</i> (<i>SD</i>)
Reexperiencing	3 (1, 5)	2.89 (2.25)	1 (0, 4)	2.00 (2.41)	2.5 (1, 4)	8.37 (7.20)
Avoidance	4 (3, 6)	4.31 (2.26)	5 (2, 6)	4.73 (2.80)	4 (2, 6)	6.48 (9.47)
Sense of threat	5 (0.25, 6)	3.69 (2.80)	3 (1.75, 7)	4.27 (2.97)	3 (2, 5.75)	2.95 (2.35)
PTSD FI	7 (0, 11)	6.46 (4.84)	2 (0, 7)	3.64 (4.30)	5 (2.25, 6.75)	4.16 (2.19)
Affect dysregulation	5 (3, 7)	5.00 (2.40)	4 (3, 7)	4.46 (2.98)	4 (3, 5)	4.11 (2.54)
Negative self-concept	4 (2.5, 7)	4.54 (2.47)	4 (0, 7)	3.73 (3.44)	5 (2, 6)	5.37 (3.06)
Difficulties in relationships	6 (3, 8)	5.27 (2.63)	3 (2, 6)	3.91 (2.59)	5 (2, 7)	4.16 (1.37)
CPTSD FI	7 (3.5, 11)	6.85 (4.04)	4 (1, 6)	4.00 (3.82)	5 (3, 7.75)	5.05 (2.59)

Note. PTSD FI = PTSD-related functional impairment; CPTSD FI = CPTSD-related functional impairment; PTSD = posttraumatic stress disorder; CPTSD = complex posttraumatic stress disorder; ITQ = International Trauma Questionnaire; IQR = interquartile ranges.

Table 4*Associations Between ITQ Symptom Clusters and Primary Drugs of Concern*

ITQ symptom cluster	Kruskal–Wallis significance test			Mann–Whitney <i>U</i> comparison ^a					
	<i>H</i>	<i>p</i>	η^2	1 versus 2		1 versus 3		2 versus 3	
				<i>U</i>	<i>r_G</i>	<i>U</i>	<i>r_G</i>	<i>U</i>	<i>r_G</i>
Reexperiencing	2.235	.327	.00	160.00	.26	361.00	.11	173.50	–.17
Avoidance	0.696	.706	–.02	217.50	.00	336.00	.14	189.50	.06
Sense of threat	0.121	.941	–.03	195.50	.00	369.50	.06	190.00	.03
PTSD FI	4.91	.086	.04	149.00	.33	305.00	.25	152.00	–.28
Affect dysregulation	2.553	.279	.01	200.00	.08	302.00	.26	186.500	.11
Negative self-concept	0.641	.726	–.02	184.00	.15	383.00	.06	194.00	–.08
Difficulties in relationships	3.779	.151	.03	135.00	.33	311.00	.21	159.50	–.16
CPTSD FI	3.989	.136	.03	142.50	.35	329.00	.19	169.50	–.19

Note. PTSD FI = PTSD-related functional impairment; CPTSD FI = CPTSD-related functional impairment; PTSD = posttraumatic stress disorder; CPTSD = complex posttraumatic stress disorder; ITQ = International Trauma Questionnaire.

^a 1 = Alcohol, 2 = cannabis, 3 = other illicit substance.

variability in the symptom profiles and type of treatment required by different substance groups. Further research is required to explore this trend. Previous research has primarily found evidence of an association between substance use and hyperarousal symptoms, although there is considerable uncertainty across results and these associations have not been examined in the context of the *ICD-11*. This study did not identify any significant relationships between recent substance use and symptoms of PTSD or CPTSD. This may be an artifact of the current sample characteristics, or low power, however, previous studies have found inconsistent results in this area—indicating that the association between current trauma-related symptoms and recent substance use is not clear. For example, Garrison-Desany et al. (2024) identified associations between recent nicotine and alcohol use and PTSD, but did not identify this association for cannabis use. Results such as these highlight the complexity of the relationship between PTSD and SUD.

Importantly, all of the CPTSD symptom clusters were found to have moderate relationships to psychological dependence, a relationship not found for any of the PTSD symptom clusters. This study was the first to examine the relationships between specific PTSD and CPTSD symptom clusters and psychological dependence, but previous research has in general indicated greater levels of comorbid

symptoms in individuals with CPTSD compared with PTSD (Maercker et al., 2022). This finding also highlights the benefits of looking at SUD beyond substance use behavior and considering in more detail the associations between trauma- and substance-related symptoms. Future research should expand on this and include measures of dependence constructs or varied measures of substance use symptoms to further elucidate associations.

The findings of this study support the notion that unique or additional treatment approaches may be needed to target-specific CPTSD symptoms (Karatzias & Cloitre, 2019; Karatzias, Murphy, et al., 2019). To date, no studies have specifically examined the treatment of comorbid SUD and CPTSD. The existing literature provides strong evidence for concurrent, trauma-focused treatments for comorbid PTSD and SUD (Roberts et al., 2022, 2023). This includes the use of integrated CBT programs utilizing Prolonged Exposure such as *Concurrent Treatment of Substance Use Disorders Using Prolonged Exposure* (Back et al., 2015). Given that programs such as this have demonstrated good effects on symptoms of PTSD and SUD in general, clinicians should consider implementing these interventions but also consider personalizing this to address potential additional treatment needs for those experiencing CPTSD symptoms. Early research indicates existing PTSD treatments may have some

Table 5*Spearman's Rank-Order Correlations of ITQ Symptom Clusters and Substance Use Characteristics*

Variable	1	2	3	4	5	6	7	8	9	10
1. Reexperiencing	—									
2. Avoidance	.52**	—								
3. Sense of threat	.52**	.42**	—							
4. Affect dysregulation	.51**	.45**	.51**	—						
5. Negative self-concept	.60**	.43**	.47**	.57**	—					
6. Difficulties in relationships	.55**	.57**	.47**	.57**	.77**	—				
7. PTSD FI	.69**	.45**	.56**	.56**	.67**	.59**	—			
8. CPTSD FI	.58**	.31*	.40**	.55**	.74**	.72**	.80**	—		
9. Psychological dependence	.07	.19	.15	.35**	.38**	.43**	.23	.43**	—	
10. Days of use	–.04	.12	.09	.09	.14	.19	.05	.13	.56**	—

Note. Correlations in bold are statistically significant. PTSD FI = PTSD-related functional impairment; CPTSD FI = CPTSD-related functional impairment; PTSD = posttraumatic stress disorder; CPTSD = complex posttraumatic stress disorder; ITQ = International Trauma Questionnaire.

* $p < .05$. ** $p < .01$.

effect on CPTSD symptoms (Karatzias, Murphy, et al., 2019) and there are existing examples of using modular treatment approaches to address symptoms of CPTSD alongside those of PTSD, such as the *Enhanced STAIR* program (Karatzias et al., 2024). In this way, clinicians should consider building on or adapting existing treatments in a deliberate way to specifically address symptoms of CPTSD.

Limitations

There are several limitations that should be considered when reviewing the current results. Primarily, the emergence of a global pandemic and the redirection of health service resources, necessitated the premature cessation of recruitment for the study. The subsequent sample size will have influenced statistical power and the generalizability of results. In particular, the small sample and unequal group sizes precluded group comparisons for the PTSD and CPTSD groups. The present study is also only able to provide screened estimates of prevalence rates for PTSD and CPTSD, as no diagnostic confirmation was obtained via clinical interview. There are, however, strong suggestions that diagnostic confirmation via clinical interview should not always be considered superior in this endeavor (Hyland & Shevlin, 2024). There remains a strong need for further research in this area. Given the complexities of collecting data in under resourced, frontline services, it is important that future research employ mechanisms to maximize the utility of data collected. The examination of a broader range of substance use characteristics, including primary substances of concern, will allow more detailed analysis of the relationships between trauma- and substance-related disorders.

Conclusion

In conclusion, the present study confirms the high prevalence of trauma exposure in individuals seeking treatment for substance use problems but also expands the literature by providing the first estimates of the rates of *ICD-11* PTSD and CPTSD in this population. These results suggest that CPTSD is a potentially important comorbid diagnosis in SUD treatment requiring further examination.

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Received September 24, 2024

Revision received February 20, 2025

Accepted April 17, 2025 ■